

Dialect levelling in England: Evidence from the *English Dialects App*

1. Introduction

Variationist research on the English of England over the past 40 years has provided ample evidence of the levelling or gradual attrition of local traditional dialect features – lexical, phonological and morphosyntactic – in favour of socially and geographically more widespread variants (see, for example, Trudgill 1983, 1986; Cheshire, Edwards and Whittle 1989; Docherty and Foulkes 1999; Britain 2009, 2010). Looking further back, however, it is clear from the traditional dialectological record that this levelling is not new. Alexander Ellis's (1889) survey of variation across the anglophone British Isles in the late 18th century lamented levelling in a number of parts of the country, for example the area around London, arguing that “the enormous congeries of persons from different parts of the kingdom and from different countries, and the generality of school education, render dialect nearly impossible” (1889, 225). Furthermore, the scholars who undertook the mid-20th-century *Survey of English Dialects (SED)* were extremely suspicious of informants who were too educated, too mobile, too urban, too young, lest their traditional dialect features be eroded (Orton *et al.* 1962-1971, 15). There is now a general consensus that the levelling of the past 150 years has been motivated, to a considerable extent, by the two major factors mentioned by Ellis above: mobility and the advent of universal education and literacy (see, further, Sayers 2009; Britain 2010, 2013, 2019; Burridge and Blaxter 2021). What both have in common, however, is uneven distribution. What Wolff made clear for mobility was and is equally true for education: “the suggestion of free and equal mobility is [...] a deception, since we don't all have the same access to the road” (1993, 253). Levelling, then, despite the apparent universality of mobility and education, is heavily conditioned by the inequalities within both.

A further concern in research on dialect levelling, and one which this chapter aims to address directly, is the fact that there has been relatively little empirical evidence of intra-regional and supra-regional levelling from multilocality studies with spatial scope – studies where the timing, data collection methods and data analysis techniques are the same or very similar, and therefore directly comparable. Many of the claims about the scope of levelling come from analyses of change in one or two sites, rather than comparative and comparable analyses across many sites. A few of the exceptions include Britain's work on the consequences of counterurbanisation

for traditional dialect in East Anglia (e.g. Britain 2014, 2020), which investigates susceptibility to levelling across a dozen both rural and urban sites, and Williams and Kerswill's (1999) research on levelling in Milton Keynes, Reading and Hull. The advent of new technologies, however, has provided alternative methods to investigate regional levelling in ways that were not possible using traditional on-the-ground fieldwork-based data collection techniques.

In this chapter, we attempt to address this by investigating dialect levelling using evidence collected from a smartphone application, the *English Dialects App (EDA)* (see Leemann, Kolly and Britain 2018; Britain, Blaxter and Leemann 2020). In the past decade, smartphone, social media and other internet-reliant technologies have been increasingly adopted to gather large datasets with a broad spatial coverage that capture language variability. These include the use of Twitter (e.g. Gonçalves and Sánchez 2014; Stevenson 2016; Grieve, Montgomery, Nini, Murakami and Guo 2019; Nini, Bailey, Guo and Grieve 2020; Willis 2020), and internet-based online surveys (MacKenzie, Bailey and Turton 2014). The affordances of such technologies include a very wide uptake from a large cross-section of the community across a wide geographical area (see also Blaxter and Britain 2021), relatively inexpensive and quick data collection, processing and analysis, as well as, especially in the case of our smartphone application, bringing language and dialect issues to a wider public in an engaging and entertaining way (Britain, Kolly and Leemann 2018). These methods, therefore, allow us to examine – at the same time and in the same way – the same variable across a very large number of people, of varied social backgrounds, living across the whole country. In the next section, we briefly introduce the *EDA* (for more detail, see Leemann *et al.* 2018). We then present the results for eight linguistic variables – two phonological, three morphosyntactic, and three lexical – which map the extent of the levelling of traditional dialect in England since the last systematic nationwide study – the *SED* (Orton *et al.* 1962-1971).

2. The English Dialects App (EDA)

Launched in early 2016, the *EDA* is a free iOS and Android application with two main functions – a dialect quiz and dialect recordings. For the quiz, app users indicate which variants of 26 words they themselves use. For the recording function, users can upload recordings of themselves reading a short version of the 'Boy Who Cried Wolf' story and can also listen to already uploaded recordings, both those submitted by other app users as well as recordings of speakers from the *SED*. Since we used the quiz data alone for the analysis of dialect levelling presented in this chapter, we say no more here about the recordings function of the app. More details can be found in Leemann *et al.* (2018) and Britain *et al.* (2018). Kirkham, Turton and Leemann (2020) and Strycharczuk, López-Ibáñez, Brown and Leemann (2020) present results from investigations of the corpus of uploaded recordings.

The aim of the quiz is to gather data on the usage of specific lexical, phonological and grammatical variables. Each screen in the quiz prompts users to either answer a question or complete a fill-the-gap task (sample screenshots can be seen in Figure 1¹). Below the questions, users can read the possible answers. For the phonological variables, users can listen to each of the possible answers as many times as they wish, by clicking on the circled arrow to the left of the answer, before choosing whichever one they think they themselves use. They are then moved to the next screen, which shows a map of where in the country their answer was also used². The app then moves on to ask the next question. Typical phonological questions can be seen in Figure 1. We also asked questions about morphosyntactic and lexical variables. Different questions had different numbers of possible answers, ranging from two to ten.

Users who wish to support our research further are directed to our ethics and privacy policy and are asked to accept the terms and conditions. If they agree, they are then asked to indicate their dialect location using a pin on a map and complete a set of social background questions on age, gender, ethnicity, mobility and education (Figure 2). In a May 2016 update to the app, an additional question about parents' education was added. Users' answers together with the social background information are then anonymously stored on a server. By the end of 2018, over 130,000 people had completed the quiz, and over 50,000 of them, from more than 4,900 localities across the UK, had additionally completed the social background metadata questionnaire on the app. In addition, more than 4,300 users had submitted audio recordings.

We are, as sociolinguists and dialectologists, fully aware of the problems and drawbacks of collecting dialect data in this way. The sample gathered with the app is, understandably, biased towards digital natives – it is younger and more educated than the population as a whole. The collection of social background data from speakers, however, means we can nevertheless control for and compare and contrast speakers' ages, educational levels, mobilities and so on. We are also aware that the quiz is not collecting dialect use but, ultimately, *intuitions* about dialect use. We present an honest account of these factors elsewhere (Leemann *et al.* 2018).

The data crowdsourced from the app quiz has enabled us to investigate a range of different features. The combination of users' location and social background information has enabled us to, for example, produce maps not only showing patterns of variation right across England (a strength of the now outdated *SED*, but a weakness of many contemporary social dialect studies of individual communities), but also consider how these patterns differ according to the age, gender, educational level and mobility of the users (a strength of locally oriented social dialect studies, but a weakness of the *SED*). We have, for example, been able to update our understanding of the location and variable nature of the 'North-South' dialect boundary (Grossenbacher, Britain,

¹ The figures are gathered at the end of the article.

² In initial versions of the app, these maps displayed the dialect distributions from the *Survey of English Dialects*; from May 2016 they showed maps based on earlier app users' answers.

Leemann, Blaxter and Kolly 2017), considering the pronunciation of the vowel in *last* – Southern /ɑ:/ versus Northern /a/ – and in *butter* – Southern /ʌ/ versus Northern /ʊ/. We have also been able to track the geographical spread of innovations, such as [f] for /θ/ in *think* (e.g. Britain *et al.* 2020), as well as those features which appear to be undergoing rapid decline, such as rhoticity (the pronunciation of non-pre-vocalic /r/ in words such as *car* and *part* (Leemann *et al.* 2018; Grossenbacher 2016). Here, however, we consider eight linguistic variables, each of which includes one regional or local traditional dialect variant that in the existing (more ‘traditional’ dialectological and/or sociolinguistic) literature either has been flagged as under threat or suffers from a lack of empirical investigation (presumably for analytical reasons such as rarity of occurrence in the timespan of a typical sociolinguistic interview, or difficulty to elicit). Of these eight, three are lexical, two are phonological and three are morphosyntactic. For each, we provide maps of the data found in the *Basic Materials* of the *SED* (Orton *et al.* 1962-1971), as well as maps drawing on the quiz results from our *EDA*. For some of the variables, to demonstrate apparent time changes, we compare older and younger app quiz respondents³. For some others, we show variant patterns that appear to be significantly affected by social constraints such as education.

We proceed on the assumption that levelling could be manifested in a number of ways:

Geographical shrinkage: the areas in which the item is used have become more restricted – in other words, some areas, which once had the variant, appear to no longer have it, even though some other areas may retain it.

Quantitative shrinkage: the areas in which the item is used may or may not have shrunk geographically, but there has been a reduction in the *extent* to which the item is used in those areas – in other words, a population at an earlier date used the item *more* than those at a later date *in the same area*.

Given our comparisons with the *SED* (data collected in the 1950s and 1960s, mostly of relatively non-mobile, old, rural men born in the final third of the 19th century), and the possibility with the *EDA* data of an apparent-time analysis, comparing older and younger speakers in the contemporary corpus, we are then able to situate in time when most of the levelling took place. Differences between the *SED* and *EDA* suggest that the most dramatic levelling has taken place sometime during the first three quarters of the 20th century. Differences between older and younger *EDA* speakers suggest ongoing levelling since the mid to late 20th century. It is worth recall-

³ All plots used to present *EDA* data show interpolation and smoothing using kernel smoothing with an adaptive square kernel (i.e. *k*-nearest neighbour smoothing) (for more on such methods see Rumpf *et al.* 2009; Sibling *et al.* 2012; Blaxter 2017, 2019). Age plots were created by weighting all the data by age using a Gaussian kernel with a bandwidth of 20 years. To estimate the usage for the maps of ‘younger’ speakers, the mean of the Gaussian curve was set at 20 years and to estimate the usage of the ‘older’ speakers, the mean was set at 80 years. Kernel smoothing in space was then applied to this weighted data in the same way for other maps.

ing, however, that the demographics of *SED* and *EDA* informants are quite different – while the *SED* sampled non-mobile, old, rural men almost exclusively, the *EDA* captured the entire community, but with a bias towards the younger, the more mobile and the more educated. When looking at the maps for different variables, therefore, it is important to remember that the *SED* maps are probably ‘best case scenario’ snapshots for local dialect (with the overall fate of traditional dialect forms at that time being *less* favourable than what those maps suggest), and the maps for the *EDA* presenting a ‘worst case scenario’ picture of traditional dialect, because those least likely to use traditional dialect are overrepresented in the *EDA* sample.

3. Evidence of lexical levelling

3.1. *Snail*

The first lexical feature we analysed here was *-dod-*containing lexemes denoting *snail*, such as *hodmedods*, *hoddy-dods*, *dodmans*, *doddermans*, etc. Although at least sporadic use of these variants of *snail* was, back in the 19th century, widespread (Wright 1906), by the time of the *SED* in the mid-20th century, as shown in Figure 3a, they were largely restricted to Eastern England, especially Norfolk, Suffolk and Essex (see also Britain 2009, Britain *et al.* 2020).

Figures 3b and 3c show the use of these forms today in the results of the *EDA* among older and younger speakers respectively. Note that the darkest shaded areas in the map show usage levels of just 10% in those areas. Among the older app users, only Suffolk and especially Norfolk witness levels of *-dod-* lexemes above 5%, and they appear to be completely absent from the areas to the west of this core area where they had been present at the time of the *SED*. Among the young their use has fallen further and become even more geographically restricted to Eastern Norfolk and parts of the far north of Suffolk. We have clear evidence here of geographical and quantitative shrinkage, and of levelling that must have already been underway in the first half of the 20th century.

3.2. *Autumn*

Here we examine the fate of the two most prominent alternative lexemes from the *SED*: *backend* and *fall*. *Fall*, the dominant form today in North America, was apparently quite widespread across England in the 19th century and *backend* was attested right across the North and the East Midlands (Wright 1906). Figures 4a and 4b show the distributions of *fall* and *backend* respectively in the *SED* from the mid-20th century. At the time of the *SED*, *fall* was found in Lincolnshire in the East, but predominantly in the South Midlands and especially the South-West. *Backend*, meanwhile, was the dominant form across the North and the Midlands.

Figures 4c and 4d show the contemporary distributions of *fall* and *backend* respectively, on the basis of the *EDA* data. *Fall* is only found these days at *extremely*

low levels – note that the areas of highest density are only using *fall* in 2% of cases. Some of these denser areas suggest apparent retentions of the form from earlier times – the area in the South-West around Bristol and the Severn estuary, for example, but also Cornwall, as well as North Buckinghamshire to the North-West of London. The cluster of instances of *fall* in Lincolnshire at the time of the *SED*, however, is no longer evident in the *EDA* data. ‘New’ regional appearances of *fall* can be seen in central London, Manchester and Sheffield on the one hand, but also, for example, in West Suffolk, perhaps not coincidentally near the home of two very large US air force bases. Clearly, though, *fall* is virtually entirely levelled, and suggests considerable geographical and quantitative shrinkage since the *SED*.

This has been the fate of *backend*, too, but to a less extreme extent. Judging by the *EDA* data, it is barely found these days in the Midlands, Lancashire, Manchester, Liverpool or in Newcastle, Sunderland and Middlesbrough in the North-East. The highest levels of use, around 10-15% of the variants used, are found in Cumbria in the far North-West and around Bridlington and Scarborough on the East coast of Yorkshire.

3.3. *Splinter*

The word used across England for a small piece of wood, etc., stuck under the skin was extremely variable at the time of the *SED*. Alongside Southern and South-Western *splinter*, we find *shiver* and *sliver* in the East (see Britain *et al.* 2020), *speel*, *spelk* and *spell* in the North and North-East, *spile* in the North-West, *splint* in the West and North-West Midlands, *speel* in Lincolnshire and Cumbria, *spill* in the West along the border with Wales, *spool* in parts of Yorkshire, as well as occasional occurrences of yet more variants (see Figure 5a). All of these forms have shown attrition since the time of the *SED* – in the *EDA*, *splinter* was the majority form everywhere, with the exception of *spelk* in the North-East, which has contracted geographically somewhat, but remains robustly used in Newcastle and its hinterland.

We focus here on the fate of *spell*, common at the time of the *SED*, as we can see in Figure 5b, in Yorkshire, parts of Northern Lancashire, Cumbria and Northern Derbyshire. Figures 5c and 5d show the geographical distribution of *spell* among older and younger users of the *EDA* respectively.

Results from the older speakers in the *EDA* – Figure 5c – show a geographical distribution very similar to that in the *SED* and, especially in West and Western North Yorkshire, levels of use at around 50% of all users. Given the extremely restricted non-mobile, older rural male sample of the *SED*, and the more mobile, more educated, less exclusively rural, male and female sample of the *EDA*, these results suggest that this form underwent relatively little attrition between the second half of the 19th century, when the *SED* informants were born, and the middle of the 20th, when our older app users were born. The picture for younger speakers, however, is quite different – Figure 5d. While the geographical area in which *spell* is used has only contracted a little – the form has been largely lost in the areas south of Yorkshire and Humberside, as well as in West Lancashire and South Cumbria, where there has been consid-

erable quantitative shrinkage – even in the areas where it is still found, it is much less commonly used than among older speakers, reaching only a maximum of 30% even in the core heartlands of West Yorkshire. Attrition for this variant, then (and unlike for other variants of *splinter* in other places, e.g. *shiver* and *sliver* in East Anglia, where obsolescence seems to have begun earlier; see Britain *et al.* 2020), appears to be a relatively recent, late-20th-century change, one that is ongoing but by no means yet near completion.

4. Evidence of phonological levelling

4.1. The CLOTH vowel

We investigate here non-standard realisations of the CLOTH vowel, exemplified in the *EDA* by the word *off*. In the 18th century we saw in Southern England the beginnings of a split of the /a/ vowel (from Middle English /a/ and /au/), with some lexemes belonging to that phoneme remaining short and front [a - æ], while others lengthened and eventually backed [æ: - a: - ɑ:]. This is today's so-called TRAP-BATH split. Many of the BATH words occur before voiceless consonants or before nasal clusters (*staff, path, pass, dance, demand*), but this environmental constraint was not watertight (*gas, hassle, cancer, ant, grand, ample* remain in the TRAP set). This split is now well-established in Southern but not Northern England. Just when the TRAP-BATH split was establishing itself, however, Middle English /ɒ/ also underwent lengthening as well as backing/raising, also only before voiceless fricatives, leading to a LOT [ɒ] - CLOTH [ɔ:] split. To complicate matters slightly, in “those parts of the South of England remote from London” (Wells 1982, 131) – parts of East Anglia and the West and South-West of England – the LOT vowel was often unrounded to [ɑ(:)], and given that the CLOTH class is a sub-set of LOT, it, too, in the relevant dialects, can be realised as an unrounded vowel. LOT, then, could be realised as [ɒ] - [ɑ] and CLOTH as [ɔ:] - [ɒ] - [ɑ(:)]. In the *EDA* quiz we enquired about the realisation of the word *off*, a member of the CLOTH lexical set.

Figure 6a shows how *off* was realised in the *SED*. Perhaps not unexpectedly, the area with lengthening e.g. to [ɔ:] (black dots in Figure 6a) or [ɑ(:)] (pink dots in Figure 6a) in the *SED* is similar to the area that, at the time, had undergone TRAP-BATH split – the South, East and South-West of England, parts of the East, West and South Midlands (the extent of [ɔ:] is actually slightly greater, reaching further into the North). The area of the unrounded vowel is relatively restricted (it showed a wider distribution for non-CLOTH members of the LOT set) to a few locations in the West of England. Figure 6b shows the distribution of [ɔ:] and Figure 6c the distribution of unrounded [ɑ(:)] for *off* in the *EDA*. Both of these two traditional forms are found concentrated in East Anglia, and the latter also in the South-West. Notice, however, that despite these concentrations, the actual rates of use are *extremely* low indeed. These forms *appear to be* on the brink of total obsolescence in England.

4.2. Clear /l/

“English” is well-known to make an allophonic distinction for /l/ between so-called ‘clear’ [l], an alveolar lateral approximant, and ‘dark’ [ɫ], a velarized version of the same approximant. Clear /l/ occurs prevocally, as in *lip* [lɪp], *leave* [li:v] and *lot* [lɒt], and dark /l/ non-prevocally as in *pill* [pɪɫ], *bolt* [bɒɫ] and *pickle* [pɪkɫ]. Not all accents of English, however, have this distinction – some have relatively velarized realisations throughout, even in prevocalic environments (e.g. Australian English), e.g. *lip* [lɪp], and some have clear forms throughout, even in phonological contexts where we might expect dark /l/ (e.g. English in the Republic of Ireland), e.g. *pill* [pɪl]. In many varieties of English today, also, non-onset, dark, velarized /l/ is undergoing vocalization (see, for example, Johnson and Britain 2007).

In England, clear forms, as shown in Figure 7a derived from data from the *SED*, were once found across the North of England and in East Anglia. Figures 7b and 7c show the geographical distribution of clear /l/ from the *EDA* quiz data, where respondents were asked to choose their own realisation from a choice of clear, dark and vocalised tokens in a non-prevocalic context (where the standard accent would expect a dark velarized [ɫ]).

The *EDA* results highlight the degree of geographical shrinkage of clear /l/ since the *SED*. Among older users of the *EDA*, clear /l/ is only found in the North-East, around Tyneside (e.g. Newcastle) and Teesside (e.g. Middlesbrough) and their rural hinterlands, as well as in Northern Cumbria in North-West England. A comparison of the older and younger users of the *EDA* shows further *geographical* shrinkage – attrition in the North-West and around Middlesbrough, but no quantitative shrinkage in the core area around Newcastle in Tyneside – indeed the younger users in Newcastle are using clear /l/ slightly more than the older users. Clear /l/, although in geographical retreat, appears to have found a relatively safe haven, for now, in the North-East.

5. Evidence of morphosyntactic levelling

5.1. Possessive pronouns: *hern*

Standard English has an irregular system for the construction of possessive pronouns (e.g. this book is *mine* – no, it’s not, it’s *hers*), sometimes taking the possessive determiner and adding *-n*, as in (1), and sometimes adding [z] (if the determiner doesn’t already end in /z/) as in (2):

(1) *my* /maɪ/ + /n/ → *mine* /maɪn/

(2) *her* /hɜ:/ + /z/ → *hers* /hɜ:z/

your /jɔ:/ + /z/ → *yours* /jɔ:z/

our /aʊə/ + /z/ → *ours* /aʊəz/

their /ðɛ:/ + /z/ → *theirs* /ðɛ:z/

cf. *his* /hɪz/ → *his* /hɪz/

Some non-standard dialects have traditionally regularised this redundant complexity and used strategy 1 above, the addition of *-n*, throughout: *mine, hern, hissen, yourn, ourn, theirn*. In our *EDA* quiz, we considered the use of *hern* for standard *hers*. Figure 8a shows the distribution of *hern* in the *SED*, and 8b its distribution in the *EDA*. In the *SED*, *hern* is found across the centre and South of England – it is largely absent from East Anglia, from the North, from the South-Western peninsular and from the area immediately around London. Figure 8b shows that *hern* has suffered both geographical and quantitative shrinkage – it is used at low levels and is almost entirely concentrated in the far North of its *SED* heartland, in central Nottinghamshire and Derbyshire, predominantly around the town of Mansfield. Even there, however, it only reaches around 13% of the responses. Around Wolverhampton in the West Midlands, there is also a weaker cluster of *hern* use.

One of the strengths of the *EDA*, as we saw earlier, is that it explicitly collected detailed social metadata about users, and this enables us to provide a *socio-geographical* perspective on attrition. In order to exemplify this, we examine the results for *hern*, looking at whether app users had no (Figure 8c), one (8d) or two (8e) parents that attended university. Across all three figures, the concentration of *hern* in the East Midlands, especially Lincolnshire, Nottinghamshire and Derbyshire, is clear, but the users whose parents did not have a university education have considerably higher rates of *hern* use than those with one university-educated parent, who in turn are more non-standard than those who had both parents having such an education. The latter group barely use the non-standard form at all. The West Midland cluster around Wolverhampton noted earlier is due almost entirely to users whose parents did not attend university.

5.2. *Reflexive pronouns*: hisself

In the case of the possessive pronouns above, we saw that Standard English had an irregular mixed system, whereas traditionally some non-standard dialects had a regularised system. The same is true for reflexive pronouns. In the standard, as we see in the table below, reflexive pronouns are formed by preceding *-self/-selves* with *either* a possessive (P) *or* an object (O) pronoun (in some cases it is ambiguous). Traditionally some non-standard dialects used possessive pronouns throughout, and this is regularly cited as a widespread feature of non-standard Englishes (e.g. Cheshire *et al.* 1989).

Person	Standard	Non-Standard
1 st singular	<i>myself</i> – P	<i>myself</i> – ambiguous ⁴
2 nd singular	<i>yourself</i> – P	<i>yourself</i> – ambiguous
3 rd singular M	<i>himself</i> – O	<i>hissel</i> – P
3 rd singular F	<i>hersel</i> – P?	<i>hersel</i> – P
3 rd singular N	<i>itsel</i> – P?	<i>itsel</i> – P
1 st plural	<i>ourselves</i> – P	<i>ourselves</i> – ambiguous
2 nd plural	<i>yourselves</i> – P	<i>yourselves</i> – ambiguous
3 rd plural	<i>themselves</i> – O	<i>theirselves</i> – P

Using the example of *hissel* vs. *himself*, the *EDA* examined the extent to which possessive rather than object prefixes are used for the reflexive pronouns. As shown in Figure 9a, virtually the whole of England uses (an array of differently suffixed) possessive *his-* affixes for the reflexive pronouns – indeed *himself* is barely attested at all by the NORMs of the *SED*. The *EDA* data in Figure 9b show that *his-* forms have become more regionally restricted, being favoured much more in the North – particularly the North-East (where 30% of responses were *his-* forms) but also West and South Yorkshire – than the South. In the South, *hissel* is the response from a much smaller number of app users, but it is noticeable that the very lowest frequencies are found in the more affluent areas of Southern England – to the South, West and North of London – while East Anglia and Eastern parts of London, the West Midlands and the South-West have, for the South, relatively higher levels of *hissel*. Non-standard possessive affixes for reflexive pronouns, then, appear to be undergoing considerable geographical and quantitative shrinkage, but considerable regional differences have emerged since the *SED* in the extent to which this form has been maintained.

5.3. Third person present-tense zero

Typologically rather unusually, Standard English retains present tense marking on regular verbs only for third person singular: *I/you/we/they feed*, but *he/she/it*

⁴ Some of these forms, in informal rapid speech, are realised phonetically in ways that make them grammatically ambiguous. Just as *my* can be realised as [maɪ], [mə], [mi], [mɔ], etc., so *myself* can also be realised with this same phonetic array of affixes. When someone utters [mi'self], therefore, is this *meself* with an object pronoun affix or a phonetic variant of *myself* with a possessive determiner affix? How would [mə'self] be interpreted, as *myself* or *meself*? Similarly, *yourself/yourself* can be realised as [jə'self]/[jə'selvz]. Since *you* and *your* can also both be realised as [jə] in rapid speech, is [jə'self] *yourself* or *youself*? *Ourselves* can be realised as [ə'selvz]. Is this *ourselves* or *usselves*? For the purposes of the investigation of the variability, therefore, it was important to choose the object/possessive contrast that was least ambiguous, hence the choice of *himself* vs. *hissel* for the *EDA*.

feeds. Other non-standard systems exist, however, in England's traditional dialects (see Rupp and Britain 2019 for an extensive discussion):

- A total lack of *-s* marking across the paradigm (typically in East Anglia).
- Variable *-s* marking across the paradigm (typically in the South-West), potentially determined, in some places, by the status of the subject as a pronoun or a full NP, with *-s* more likely with NP subjects and non-adjacent pronouns (the so-called *Northern Subject Rule*).
- The use of periphrastic *do* forms, such as *do feed* (typically in the South-West, potentially with somewhat different semantics to non-standard forms) (see Godfrey and Tagliamonte 1999).

In the *EDA* we examined responses to the following fill-the-gap task: *Every day on her walk past the lake she _____ the ducks (do feed? / feed? / feeds?)*. We focus here on the use of non-standard zero forms – *she feed*.

Figure 10a shows the distribution of zero forms in the *SED*. Zero forms are found at least variably in several parts of the country at low levels – across the southernmost parts of the North – Lincolnshire, Derbyshire, Cheshire – as well as the South Midlands, with higher concentrations in East Anglia, especially Norfolk and Suffolk, on the one hand, and the South-West, especially Somerset, Dorset and Cornwall, on the other. Figure 10b shows the results for older speakers in the *EDA*. The highest rates of zero (at only 10-12% maximum) are found in non-Fenland Norfolk and Suffolk, with another but lower concentration in Cornwall in the South-West. Figure 10c considers the younger users of the app. The results are partly as expected, but partly also intriguing. On the one hand, the (admittedly low %) concentration of zero forms in the same parts of East Anglia and Cornwall is clearly visible. The quantitative shrinkage that we see in East Anglia has been confirmed by more traditional variationist sociolinguistic analyses of the same feature (see Potter 2018, Rupp and Britain 2019). On the other hand, and unlike among the older app users, there are other small areas where usage levels reach well above 5% – around Preston in Lancashire, Birmingham, Wolverhampton, Rotherham in South Yorkshire, Manchester, Oldham, as well as in the far North near the Scottish border. These smaller ‘hotspots’ were not seen among the older app users, and warrant further investigation.

6. Conclusion

We have seen in the analyses of these eight variables that all have undergone geographical shrinkage, quantitative shrinkage, or both, in some cases to such an extent that the survival of the variant as a systematic local dialect feature appears questionable, even in the near future. We have also seen, furthermore, firstly, examples of obsolescing variants that show considerably more attrition among app users who have been socialised in households with university-educated parents than among those who were not, and secondly that, in a number of cases, the surviving variants have retreated to the *relatively* less mobile, less populated, more peripheral areas of

the country – to North-East East Anglia, for example, to the South-West, to the far North-West – and none have survived in good health in the highly mobile South-East. Both education and mobility, then, as Ellis warned back in the 19th century, remain key factors in helping us understand the social mechanics behind levelling.

Nevertheless, while attrition is clearly in action here, the results seem to suggest that this has affected different variants in different places in different ways to different extents, reemphasising the importance of seeing variants such as these as levelling rather than levelled. It is also important to note that our investigations of some other traditional dialect forms in the *EDA* have found relative stability over time. Despite the much younger, more educated, more mobile sample of the *EDA*, dative alternation variants such as *she gave it me* (versus *she gave me it* and *she gave it to me*) and the realisation of /ŋ/ in stressed syllables (e.g. in *sing*, *wrong*) as /ŋg/ appear not to be obsolescing at all. The picture is not entirely bleak for the traditional dialect forms, therefore, and each variant has a different story to tell.

Another conclusion that we can surely draw, however, is the value of the empirical affordances of using smartphone and internet-based technologies to gather data of this kind. The app has enabled us not just to collect data from very large numbers of people – numbers that would be inconceivable using traditional data-gathering techniques – but it can do this quickly, relatively cheaply and in ways that make subsequent data management and analysis far, far quicker too. And despite the skewing of the sample, the collection of the social metadata enables us to control, at least partially, for a range of social background influences on variant use that can overcome this skewing (see Leemann *et al.* 2018). The app provides a window on variability at the local, regional and national level, a perspective that has not been possible since the *SED* over a half century ago. It enables us to localise where variants are under threat and where they are healthy, and enables us to take into consideration who in these locations is more or less advanced in the process of attrition. It also makes possible the investigation of variants that occur only relatively sparsely within the confines of the typically hour-long sociolinguistic interview, and which would therefore probably go un(der)investigated. Results from studies such as this one can undoubtedly provide vital initial information, furthermore, to researchers wishing to drill down in particular locations to explore local variability in more detail and with more social subtlety. Just as the rapid anonymous survey produced such useful low-cost, high-yield starting points both initially for Labov (1972) in his investigations in New York and subsequently for many other social dialectologists since, so the use of social media technologies have the potential to gather critical foundational data that can act as a springboard for regional dialectologists and sociolinguists alike.

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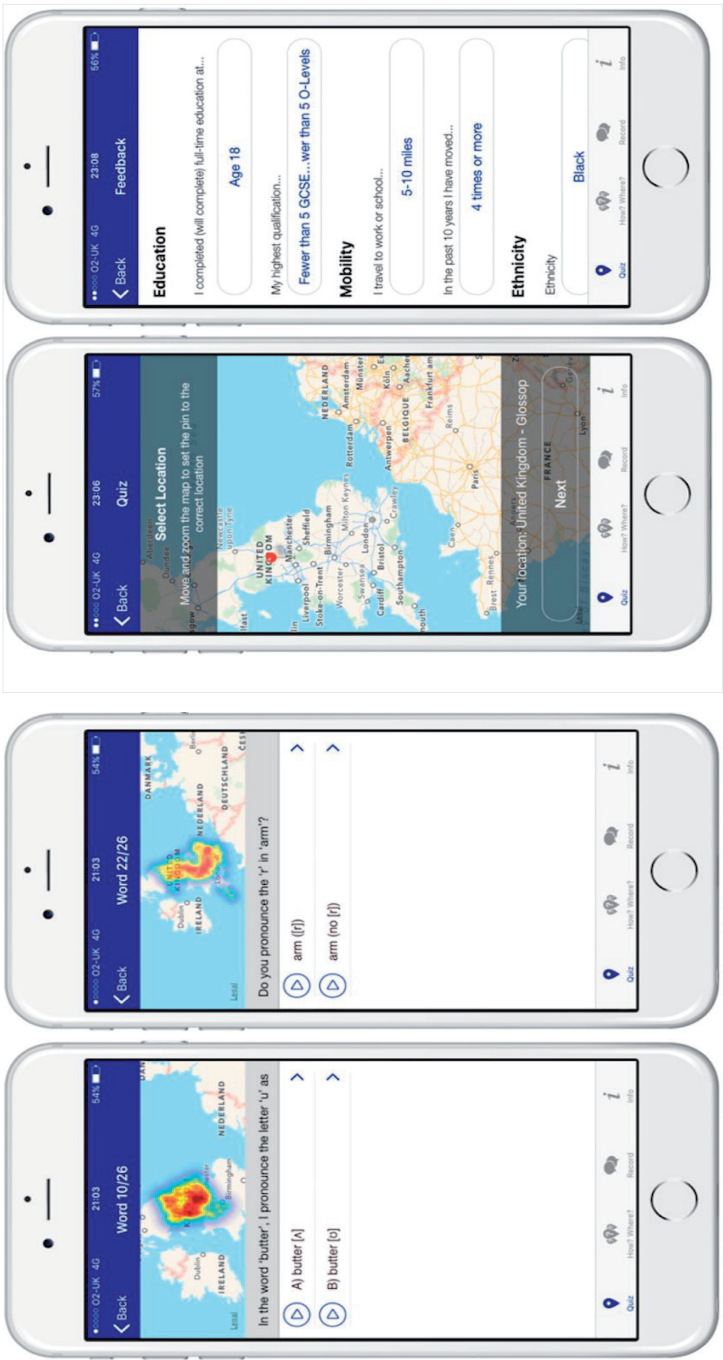


Figure 1: Prompts for variant selection in the EDA quiz

Figure 2: The app prompts users to place a pin on the locality that best corresponds to their dialect (left), and prompts them to provide social background metadata (right)

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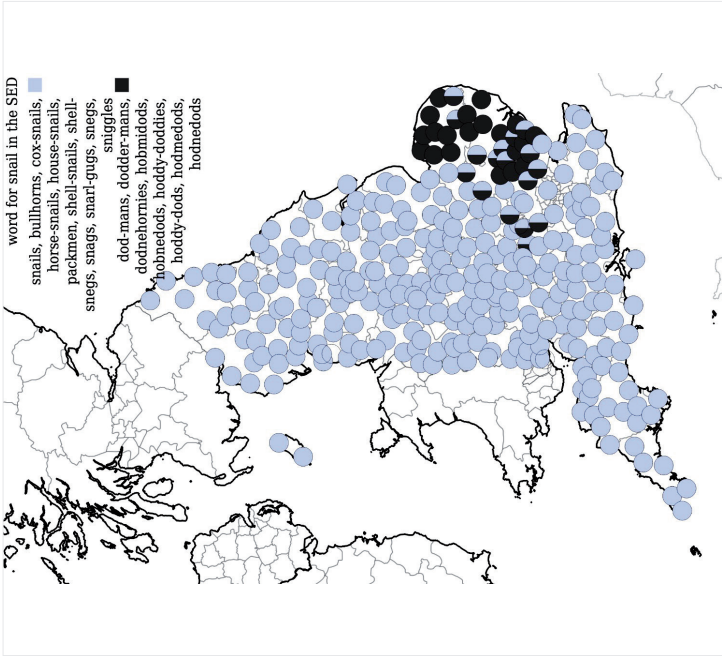


Figure 3a: The geographical distribution of *-dod*-containing lexemes for *snail* in the Survey of English Dialects (Orton *et al.* 1962-71)

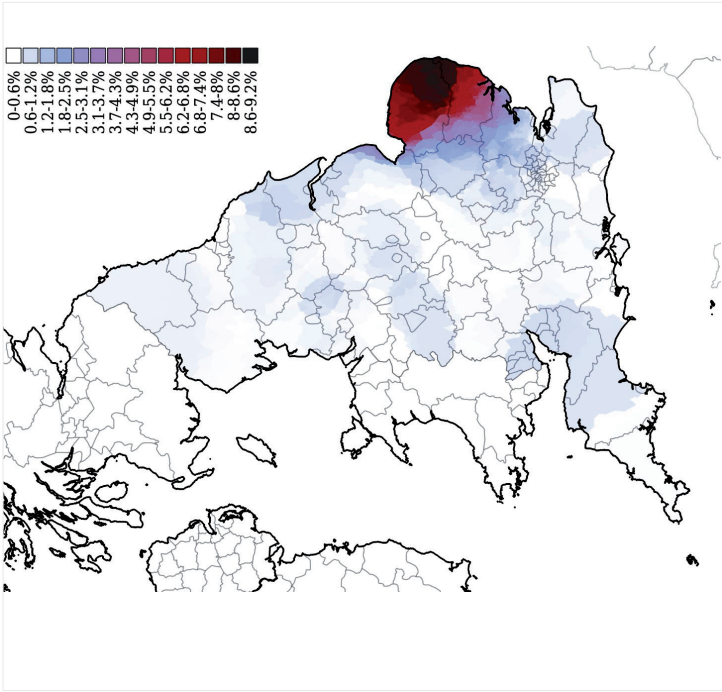


Figure 3b: The geographical distribution and degree of use of *-dod*-containing lexemes for *snail* among older users of the *English Dialects App*

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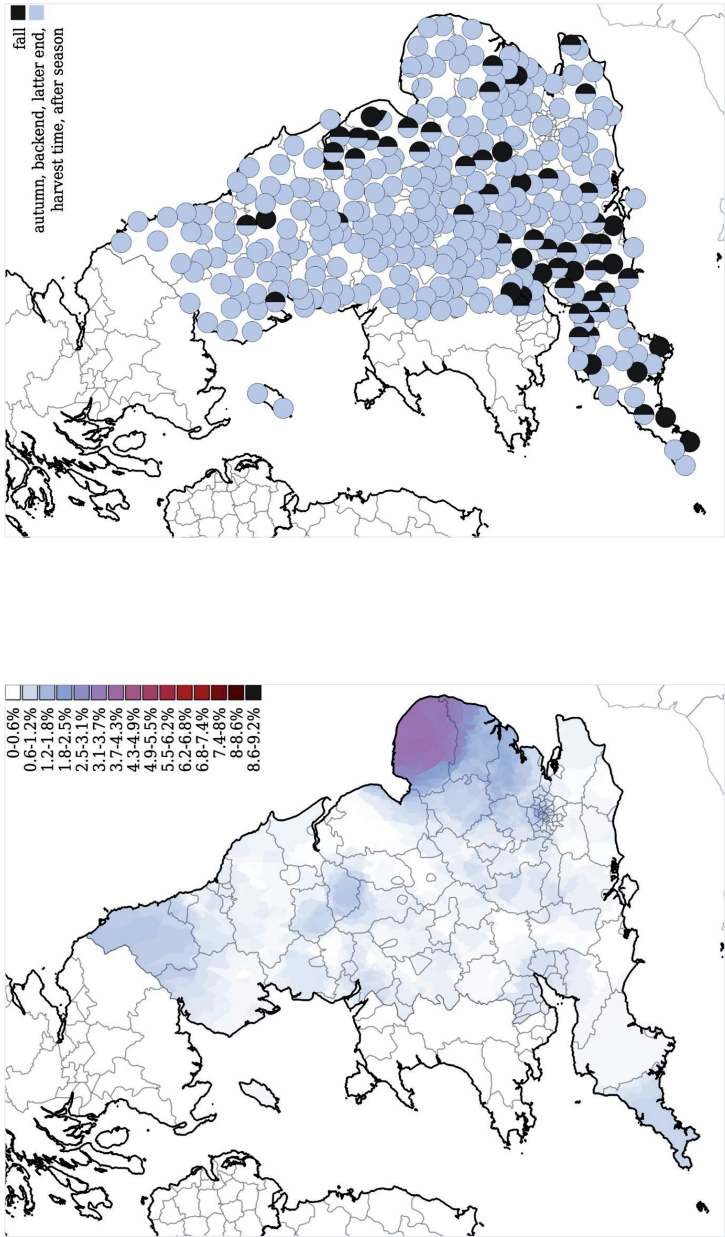


Figure 4a: The geographical distribution of *fall* for *autumn* in the *Survey of English Dialects* (Orton et al 1962-71).

Figure 3c: The geographical distribution and degree of use of *-dod*-containing lexemes for *snail* among younger users of the *English Dialects App*

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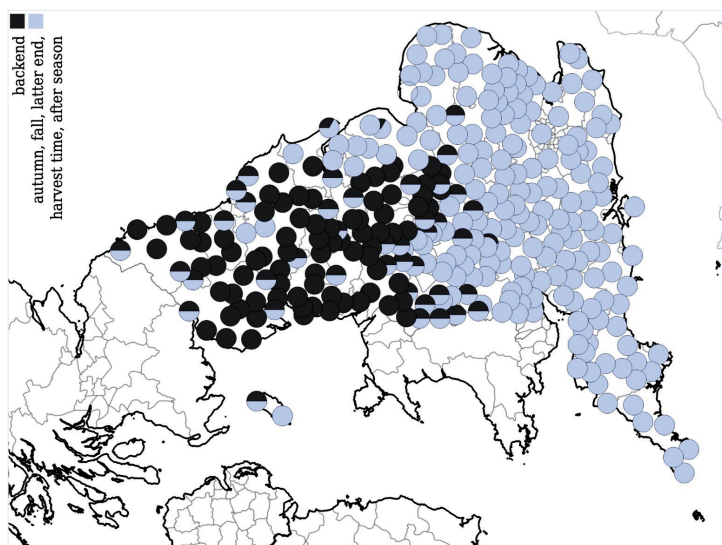


Figure 4b: The geographical distribution of *backend* for *autumn* in the Survey of English Dialects (Orton et al 1962-71).

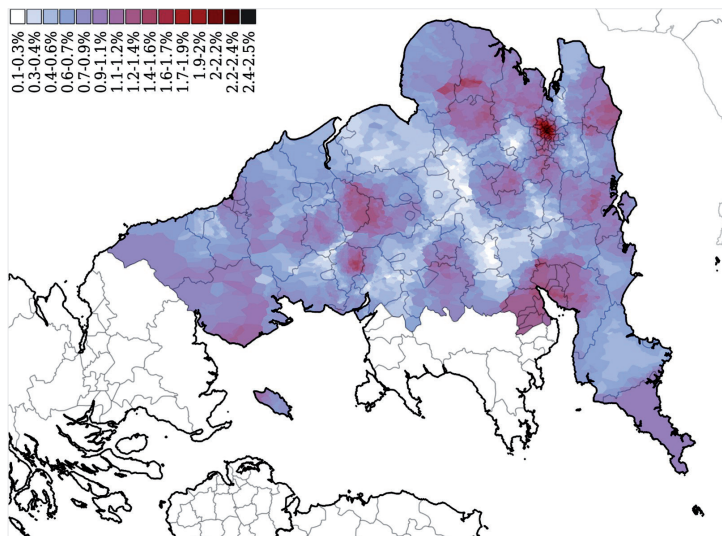


Figure 4c: The geographical distribution of *fall* for *autumn* in the English Dialects App.

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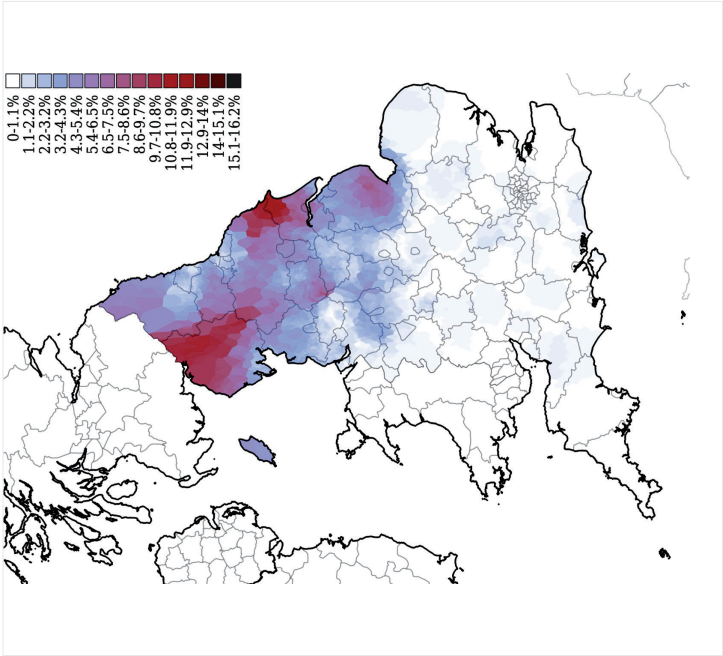


Figure 4d: The geographical distribution of *backend* for *autumn* in the *English Dialects App*.

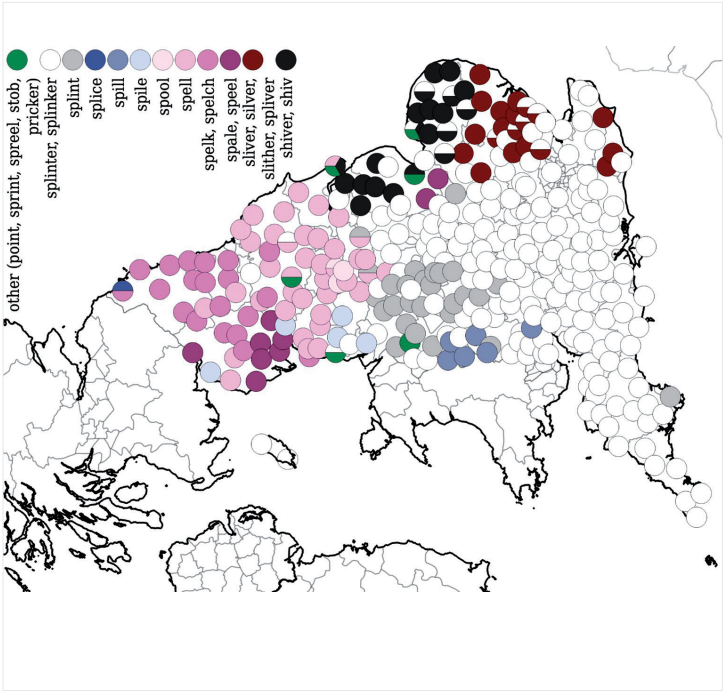


Figure 5a: The geographical distribution of variants of *splinter* in the *Survey of English Dialects* (Orton et al 1962-71).

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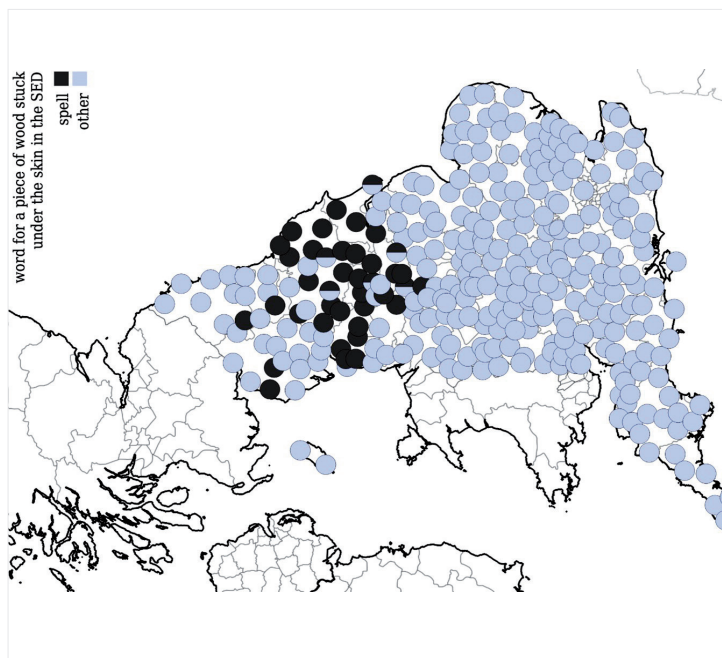


Figure 5b: The geographical distribution of *spell* for *splinter* in the Survey of *English Dialects* by locality (Orton et al 1962-71).

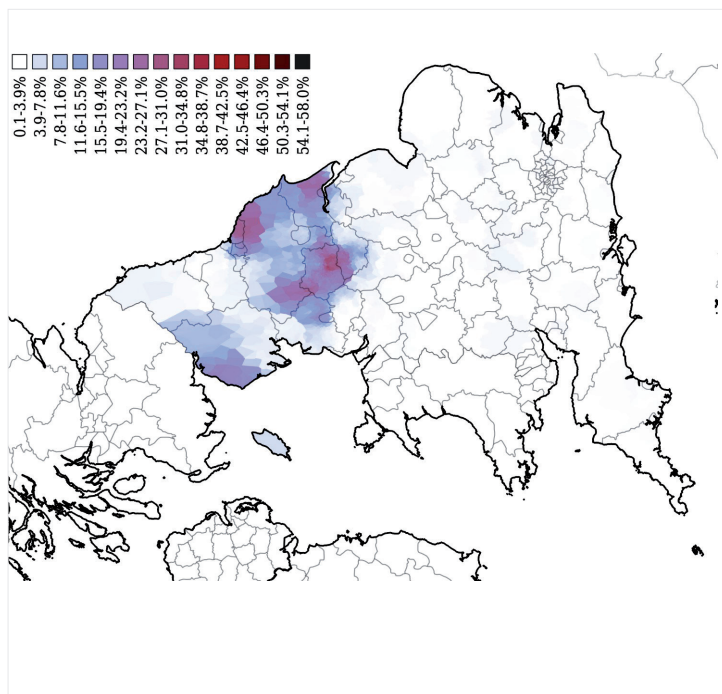


Figure 5c: The geographical distribution of *spell* for *splinter* among older speakers in the *English Dialects App*.

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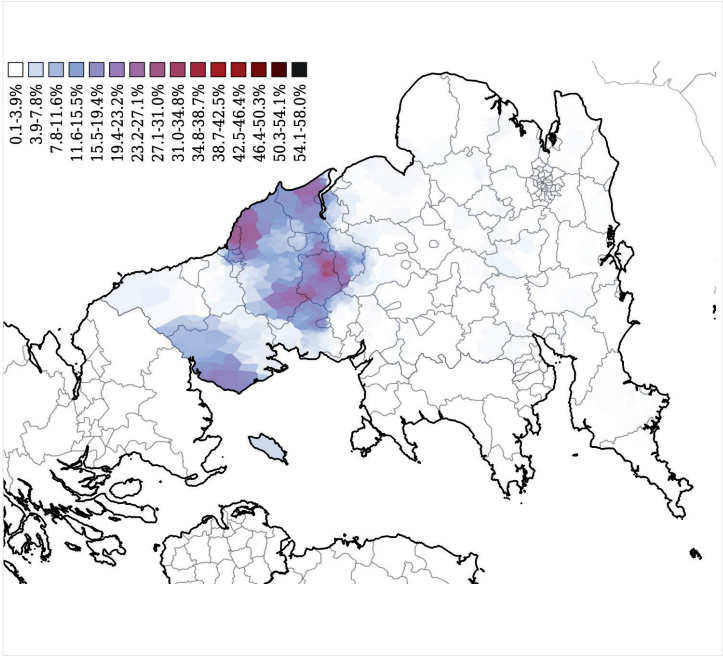


Figure 5d: The geographical distribution of *spell* for *splinter* among younger speakers in the *English Dialects App*.

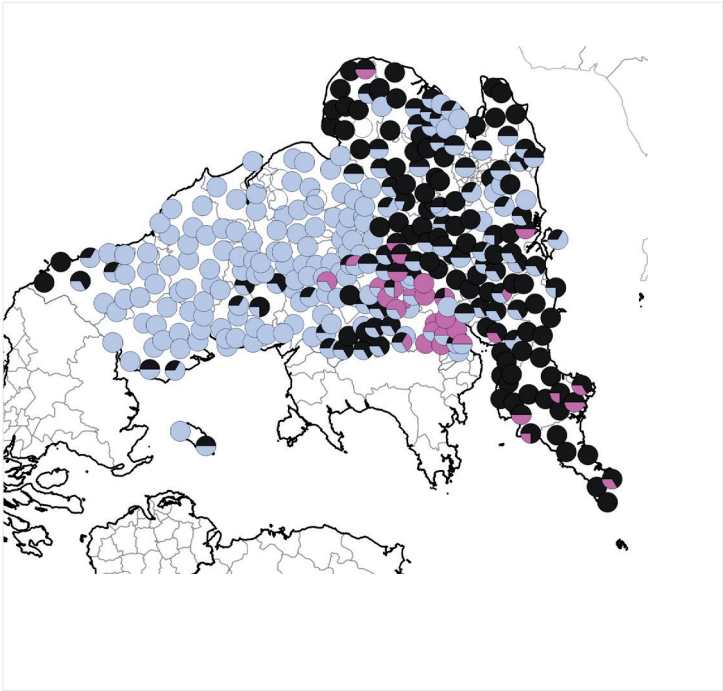


Figure 6a: *Off* in the *Survey of English Dialects*: Black = long rounded, Pink = long unrounded, Blue = short rounded.

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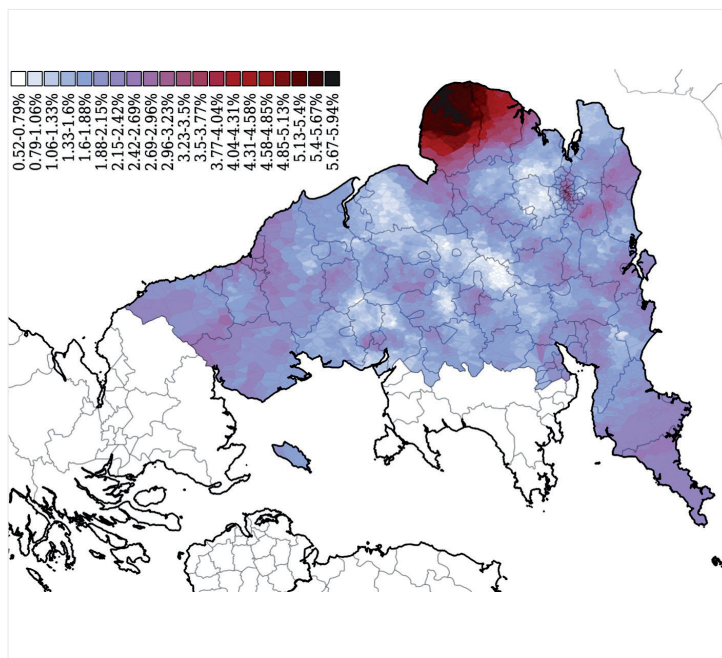


Figure 6b: *Off* as [ɔ] in the *English Dialects App*.

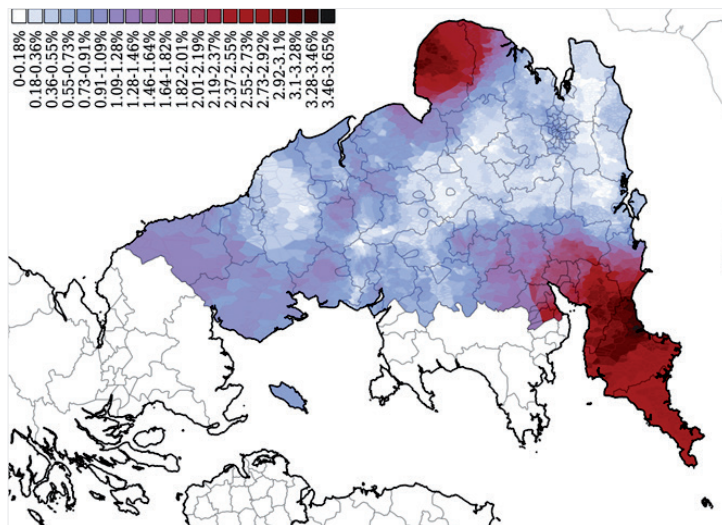


Figure 6c: *Off* as [ɑ] in the *English Dialects App*.

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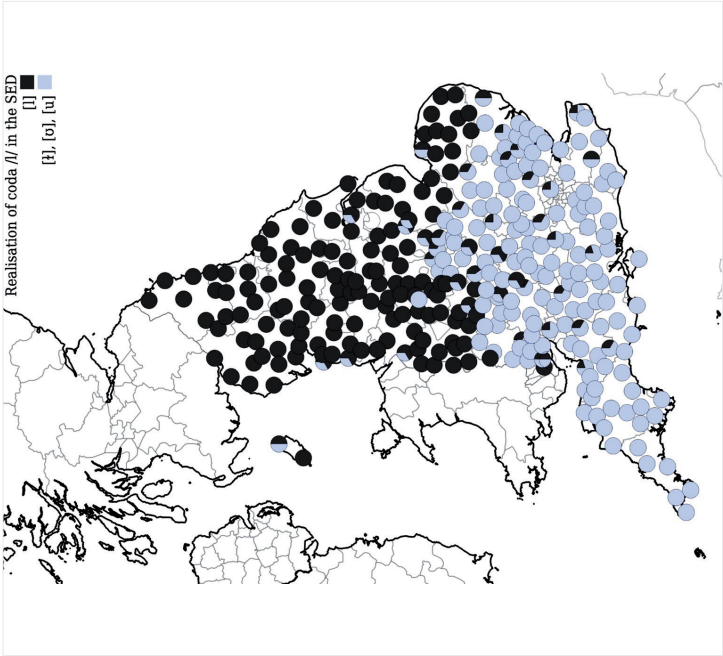


Figure 7a: The geographical distribution of 'clear' /l/ = [l] in the *Survey of English Dialects* (Orton et al 1962-71).

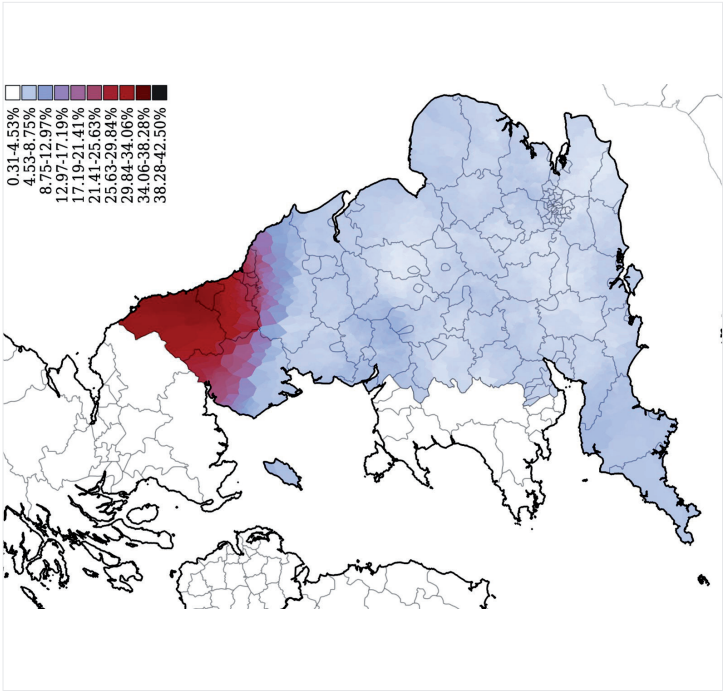


Figure 7b: The geographical distribution and degree of use of clear /l/ = [l] among older users of the *English Dialects App*

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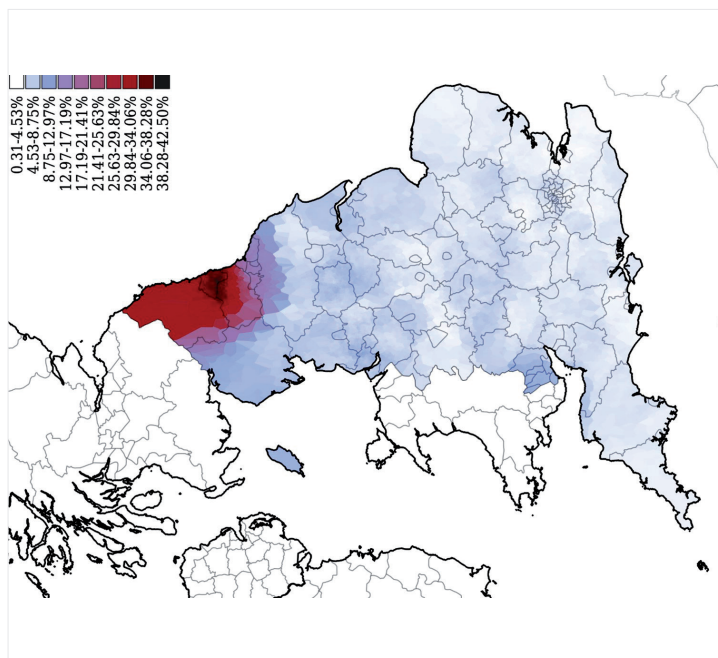


Figure 7c: The geographical distribution and degree of use of clear /l/ = [l] among younger users of the *English Dialects App*

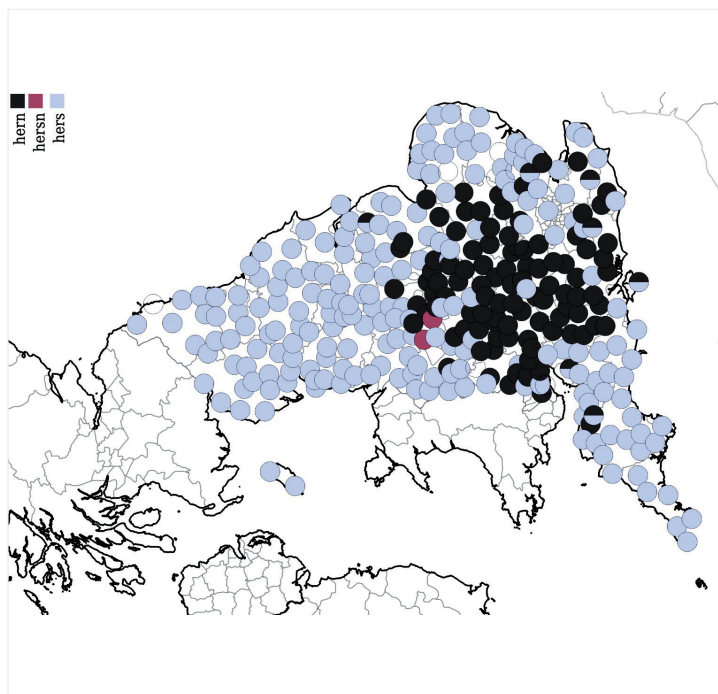


Figure 8a: The geographical distribution of *hern* for standard *hers* in the *Survey of English Dialects* (Orton et al 1962-71).

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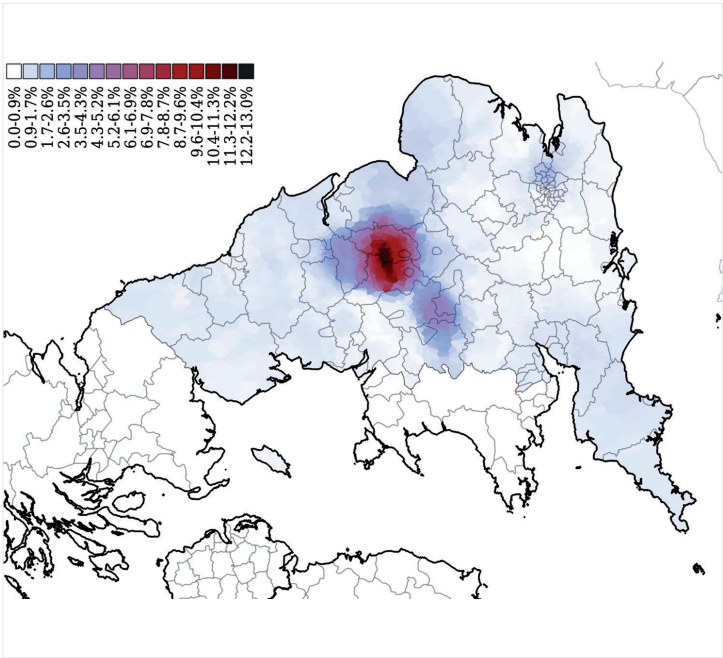


Figure 8b: The geographical distribution of *hern* for standard *hers* in the *English Dialects App*

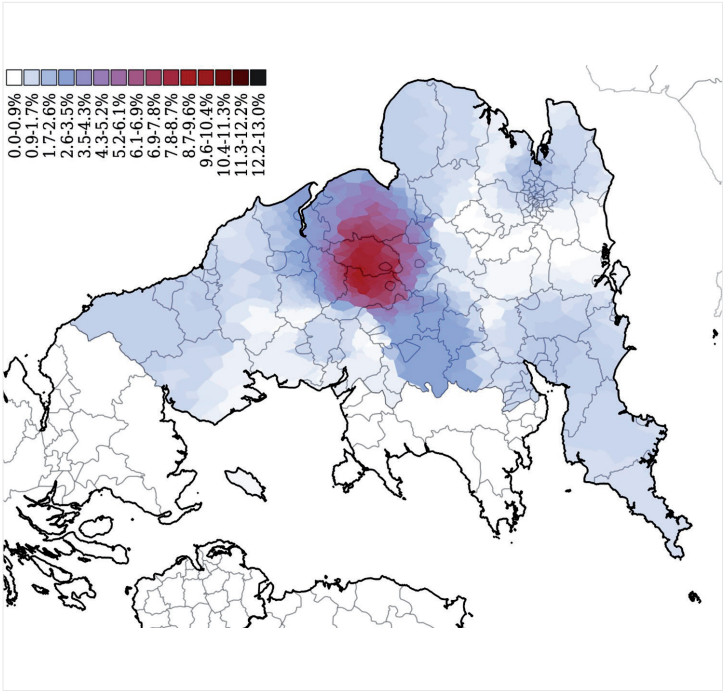


Figure 8c: The geographical distribution of *hern* for standard *hers* in the *English Dialects App* among users who had no parents that had attended university.

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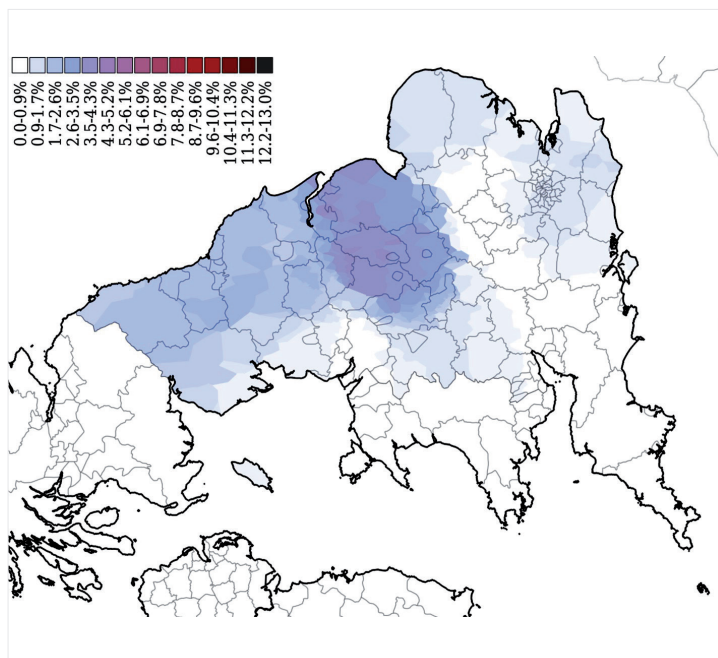


Figure 8d: The geographical distribution of *hern* for standard *hers* in the *English Dialects App* among users who had one parent that had attended university.

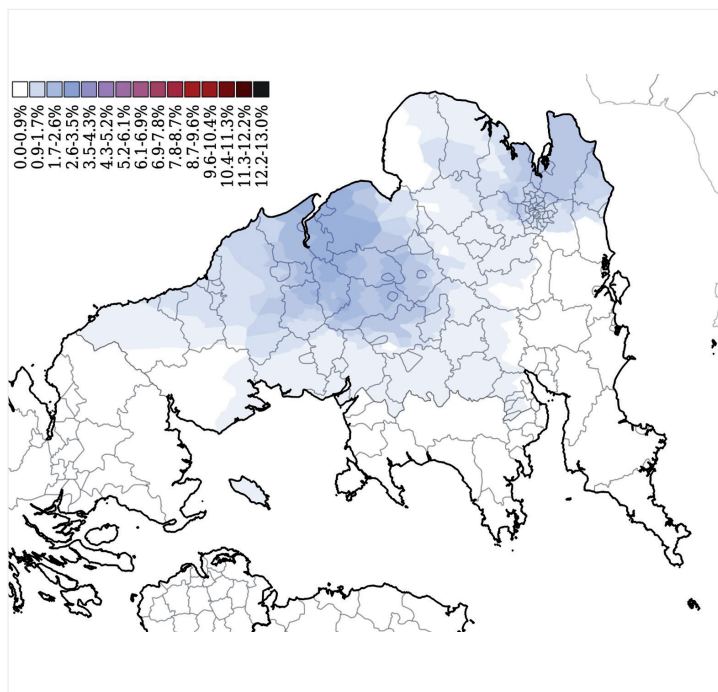


Figure 8e: The geographical distribution of *hern* for standard *hers* in the *English Dialects App* among users both of whose parents had attended university.

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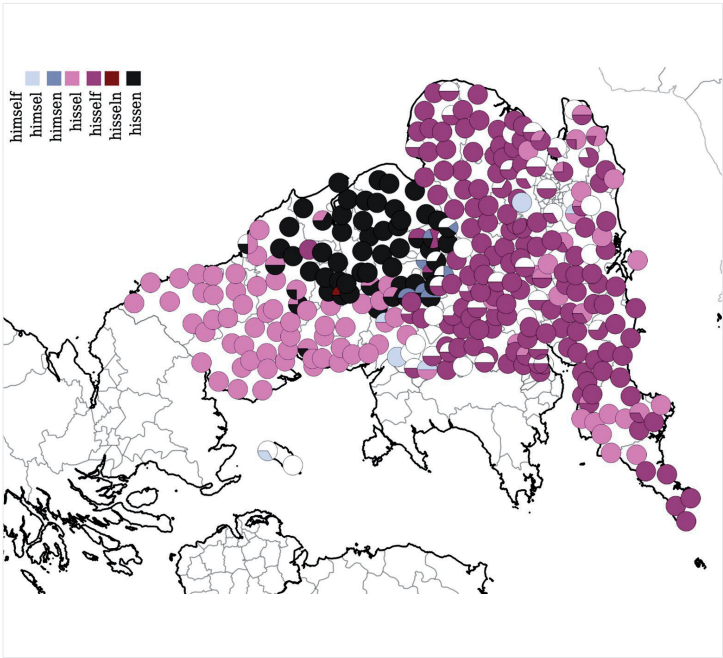


Figure 9a: The geographical distribution of non-standard possessive (as opposed to standard object) affixes of reflexive pronouns in the *Survey of English Dialects* (Orton et al 1962-71): *hisself*

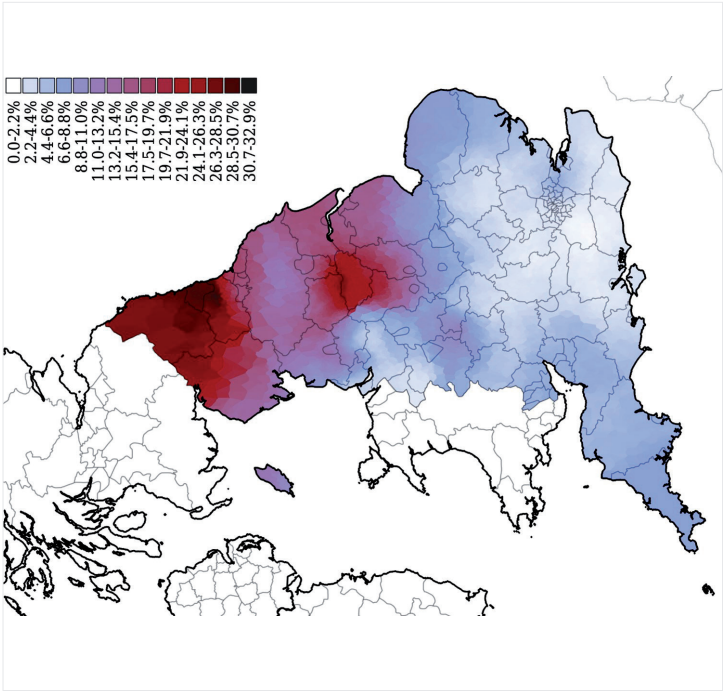


Figure 9b: The geographical distribution of non-standard possessive (as opposed to standard object) affixes of reflexive pronouns in the *English Dialects App*: *hisself*

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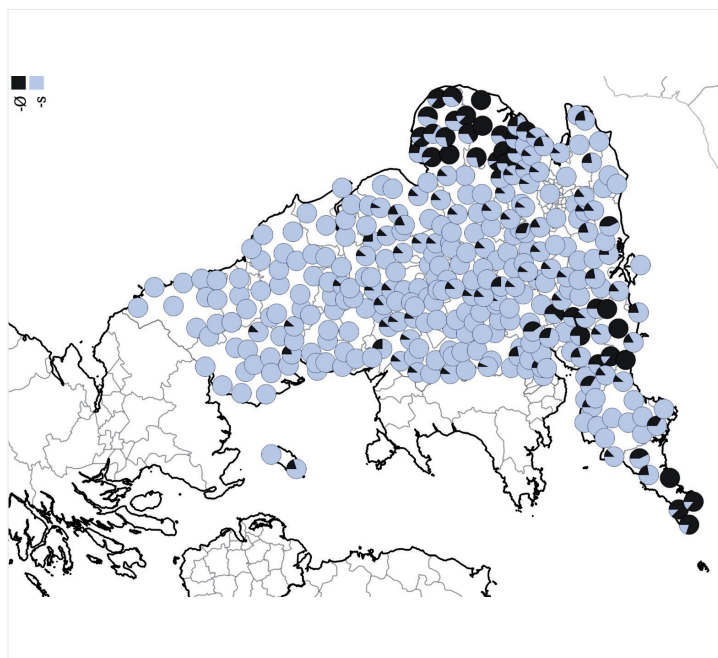


Figure 10a: The geographical distribution of third person present tense zero forms in the *Survey of English Dialects* (Orton et al 1962-71): *she feed*

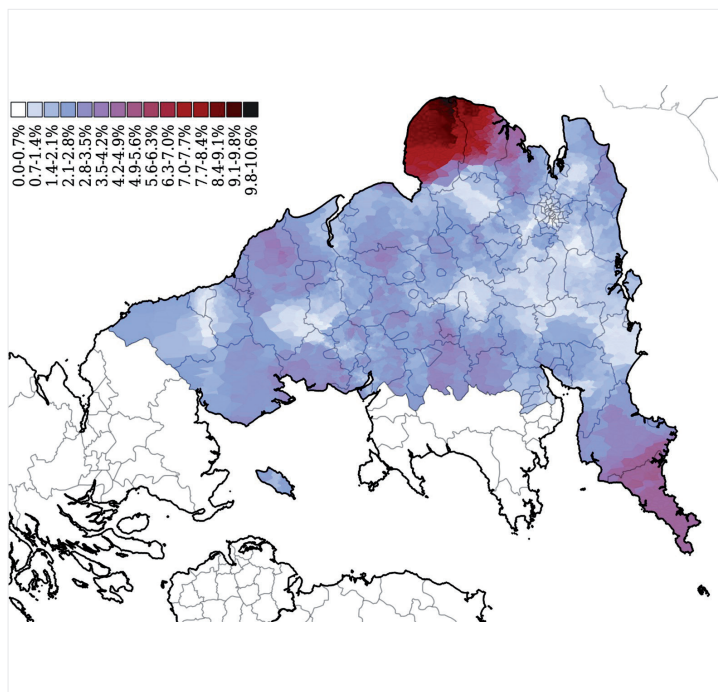


Figure 10b: The geographical distribution of third person present tense zero forms among older users of the English Dialects App: *she feed*

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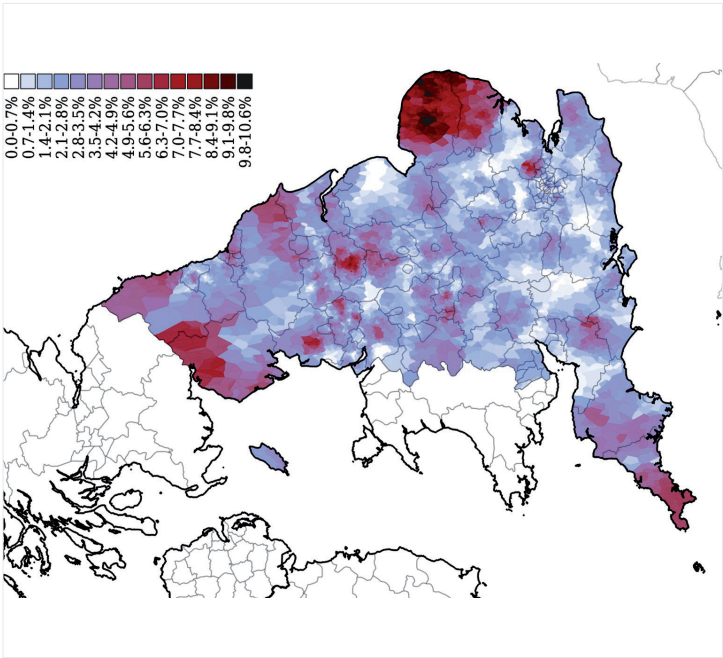


Figure 10c: The geographical distribution of third person present tense zero forms among younger users of the English Dialects App: *she feed*